

Dynamic Macroeconomics II

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Fiscal Policies in the Growth Model

First Part

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- Analyze implications of fiscal policy on choices.
- Here we take fiscal variables as **exogenous**.
- In the next topic we will calculate **optimal** fiscal policy.

- Non-stochastic model with distorting taxes and exogenous government spending.
- Sequence of government purchases $\{g_t\}_{t=0}^{\infty}$ and taxes $\{\tau_{ct}, \tau_{kt}, \tau_{nt}, \tau_{ht}\}_{t=0}^{\infty}$ on consumption, capital earnings, labor earnings, and lump-sum, respectively. There is also a sequence of τ_{it} , a subsidy on investment.
- Distorting taxes prevent a competitive equilibrium from solving a Planner's Problem.
- First Welfare Theorem does not hold. Need to solve for Competitive Equilibrium.

- Utility function over consumption and leisure:

$$\sum_{t=0}^{\infty} \beta^t U(c_t, 1 - n_t) \quad 0 < \beta < 1$$

- U is strictly increasing in c_t and $1 - n_t$, twice continuously differentiable and strictly concave.
- Firms produce the only good in the economy using a homogeneous of degree one production function with positive and decreasing marginal products of capital and labor:

$$y_t = F(k_t, n_t)$$

- Law of motion for capital:

$$k_{t+1} = (1 - \delta)k_t + x_t$$

where x_t is gross investment.

- Feasibility condition:

$$y_t = c_t + x_t + g_t$$

Competitive equilibrium

- We solve for it assuming all trades take place at $t = 0$.
- We know that this is equivalent to solving the model sequentially.
- If there was uncertainty we would be talking of solving the Arrow-Debreu equilibrium.

- **Definition: Price system:** triple of sequences $\{q_t, r_t, w_t\}_{t=0}^{\infty}$.
- q_t : time $t = 0$ pretax price of one unit of investment or consumption at time t .
- r_t is also an intertemporal price. I.e. the time $t = 0$ price of a unit of capital.
- So is w_t .

- **Definition: Budget feasible fiscal policy (BFFP):**
- Sequences of government spending, taxes and subsidy that satisfy (**intertemporal**) government budget constraint.
- The set of competitive equilibria is indexed by alternative BFFP.

- Government's intertemporal budget constraint:

$$\sum_{t=0}^{\infty} q_t g_t \leq \sum_{t=0}^{\infty} \{ \tau_{ct} q_t c_t - \tau_{it} q_t [k_{t+1} - (1 - \delta) k_t] \\ + r_t \tau_{kt} k_t + w_t \tau_{nt} n_t + q_t \tau_{ht} \}.$$

- How do we construct it?
- Start from seq. constraint at $t = 0$. Assume initial level of government assets is 0. Impose a Transversality Condition.

- Consumer's intertemporal budget constraint:

$$\begin{aligned} & \sum_{t=0}^{\infty} \{q_t(1 + \tau_{ct})c_t + (1 - \tau_{it})q_t[k_{t+1} - (1 - \delta)k_t]\} \\ & \leq \sum_{t=0}^{\infty} \{r_t(1 - \tau_{kt})k_t + w_t(1 - \tau_{nt})n_t - q_t\tau_{ht}\}. \end{aligned}$$

- How do we construct it? Analogously.

- Household chooses $\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}$ to maximize utility subject to her intertemporal budget constraint.
- Firms choose $\{k_t, n_t\}_{t=0}^{\infty}$ to maximize:

$$\sum_{t=0}^{\infty} [q_t F(k_t, n_t) - r_t k_t - w_t n_t].$$

Definition: Competitive equilibrium with distortionary taxes (CEDT):

A budget-feasible government policy, a feasible allocation for the consumer (satisfies feasibility), and a price system such that, given the price system and the government policy, the allocation solves the household's problem and the firm's problem.

- **No arbitrage:** rewrite household's budget constraint collecting terms with k_t to get

$$\begin{aligned} & \sum_{t=0}^{\infty} q_t(1 + \tau_{ct})c_t \\ \leq & \sum_{t=0}^{\infty} w_t(1 - \tau_{nt})n_t - \sum_{t=0}^{\infty} q_t\tau_{ht} + \\ & \sum_{t=1}^{\infty} [r_t(1 - \tau_{kt}) + (1 - \tau_{it})q_t(1 - \delta) - q_{t-1}(1 - \tau_{it-1})]k_t + \\ & [r_0(1 - \tau_{k0}) + (1 - \tau_{i0})q_0(1 - \delta)]k_0 - \lim_{T \rightarrow \infty} (1 - \tau_{iT})q_T k_{T+1} \\ = & W - T + R + R0 - L \end{aligned}$$

- What decision would consumer make regarding k_t ?

$$k_{t+1} = (1 - \delta)k_t + i$$

- In an equilibrium, the right side of the previous equation must be bounded.
- Consider a component of R :

$$[r_t(1 - \tau_{kt}) + (1 - \tau_{it})q_t(1 - \delta) - q_{t-1}(1 - \tau_{it-1})]k_t.$$

- If term in brackets $[...] > 0$ then consumer would choose a large $k_t > 0$ to finance unbounded consumption.
- We would have $r_t(1 - \tau_{kt}) + (1 - \tau_{it})q_t(1 - \delta) > q_{t-1}(1 - \tau_{it-1})$
- Consumer would buy capital cheap in $t - 1$, sell it expensive in t , make a profit.

- On the other hand, if term in brackets [...] < 0 then consumer would choose a negative amount $k_t < 0$ to finance unbounded consumption.
- We would have $r_t(1 - \tau_{kt}) + (1 - \tau_{it})q_t(1 - \delta) < q_{t-1}(1 - \tau_{it-1})$
- Households can *short sell*. I.e borrow capital, sell it a high price in $t - 1$, repurchase it in t at a low price, return it, make a profit.
- Note that up to this point we have not imposed $k_t \geq 0$.

- Therefore, for all $t \geq 1$, we get the No Arbitrage Condition (NAC):

$$[r_t(1 - \tau_{kt}) + (1 - \tau_{it})q_t(1 - \delta) - q_{t-1}(1 - \tau_{it-1})] = 0.$$

- This equation looks like an equation you know. **Which one?**
- Let us think about term R_0 in budget constraint.
- At $t = 0$ term in brackets can have any sign. k_0 is exogenous.

- What about term $-L$ in the budget constraint?

$$- \lim_{T \rightarrow \infty} (1 - \tau_{iT}) q_T k_{T+1}$$

- Consumer would not choose $k_{T+1} > 0$ as that would reduce its consumption possibilities.
- Consumer would like to choose $k_{T+1} < 0$.
- But as $T \rightarrow \infty$ nobody would be willing to finance that lending.

- We impose the optimality condition

$$\lim_{T \rightarrow \infty} (1 - \tau_{iT}) q_T k_{T+1} = 0$$

- Getting ahead of ourselves: q_t is linked to the marginal utility of consumption.
- Taking that into account, this limit is the Transversality Condition for this economy.

- Another formula: **user cost of capital (UCC)**
- Rewrite the NAC as:

$$r_{t+1} = \left(\frac{1}{1 - \tau_{k+1}} \right) [q_t(1 - \tau_{it}) - q_{t+1}(1 - \tau_{it+1}) + \delta q_{t+1}(1 - \tau_{it+1})]$$

- Let us assume that firms choose how much capital to accumulate.
- This formula tells us the costs involved in owning a unit of capital, instead of selling it.
- Think of a businessman deciding what to do with a plant: keep or sell.

Effects of exogenous variables:

- Higher tax implies higher capital is more expensive to operate.
- A higher depreciation rate increases the cost of capital.
- An increase in the price of capital *reduces* the cost.
- Subsidy to investment reduces cost, in steady state.

Problem of consumer is to maximize intertemporal utility subject to the (unique) intertemporal constraint, with multiplier μ .
First-order conditions (**check**):

$$\beta^t U_{1t} = \mu q_t (1 + \tau_{ct})$$

$$\beta^t U_{2t} = \mu w_t (1 - \tau_{nt})$$

- Firms solve:

$$\sum_{t=0}^{\infty} [q_t F(k_t, n_t) - r_t k_t - w_t n_t].$$

- Applying Euler's theorem on homogenous functions of degree one to $F(k, n)$, this is equivalent to:

$$\sum_{t=0}^{\infty} [(q_t F_{kt} - r_t) k_t - (q_t F_{nt} - w_t) n_t].$$



- Assume free entry. Zero Profit conditions are:

$$r_t = q_t F_{kt}$$

$$w_t = q_t F_{nt}.$$

An equilibrium is:

- A price system
- A budget feasible fiscal policy
- An allocation

that solve the system of nonlinear difference equations composed by

- Feasibility
- NAC
- Consumer's first-order conditions
- Zero Profit conditions

with k_0 and Transversality Condition (which is a terminal condition) given.

A particular case

- For simplicity, labor supply is exogenous.
- Preferences: $U(c, 1 - n) = u(c)$, $n = 1$.
- Therefore,

$$U_{1t} = u'(c_t)$$

- Define

$$f(k) = F(k, 1).$$

which implies

$$\begin{aligned} F_k &= f'(k) \\ F_n &= f(k) - f'(k)k. \end{aligned}$$

- Feasibility condition becomes:

$$f(k_t) = c_t + g_t + k_{t+1} - (1 - \delta)k_t.$$

Obtaining a second order difference equation for capital

- Iterate one-period forward no-arbitrage condition:

$$q_t(1 - \tau_{it}) = r_{t+1}(1 - \tau_{kt+1}) + (1 - \tau_{it+1})q_{t+1}(1 - \delta)$$

- Note that

$$q_t = \frac{\beta^t u'(c_t)}{\mu(1 + \tau_{ct})}.$$

Substitute prices into Euler equation to obtain (**check**):

$$u'(c_t) = \beta u'(c_{t+1}) \frac{1 + \tau_{ct}}{1 + \tau_{ct+1}} \left[\frac{1 - \tau_{kt+1}}{1 - \tau_{it}} f'(k_{t+1}) + \frac{1 - \tau_{it+1}}{1 - \tau_{it}} (1 - \delta) \right],$$

- Use feasibility condition to replace consumption in previous expression and get second-order, non-linear, difference equation.
- Initial condition is k_0 , terminal condition given by Transversality.